AI PRE-PLACEMENT TRAINER

AI Interview Module — Industry-Grade Implementation Guide

Complete Bug Analysis • Code Review • Fix Implementation Plan

Document Type
Technical Implementation Guide — AI Interview Module Only
Code Reviewed
Voice Mode ZIP (AI-BASED-TRAINER-FEEDBACK-MODEL_voice_mode-main)
Bugs Found
14 Issues: 6 Critical, 4 High, 3 Medium, 1 Low
New Issues
5 additional issues not in Blueprint v1 — found in actual voice-mode code
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1. EXECUTIVE SUMMARY — CODE REVIEW FINDINGS

This document is based on a FULL LINE-BY-LINE review of every interview-related file in the AI-BASED-TRAINER-FEEDBACK-MODEL_voice_mode-main ZIP. Unlike Blueprint v1 which analysed an older version, this review covers the actual voice-mode code that is currently in the repository.

1.1 What Changed Since Blueprint v1

✅ The voice_mode ZIP is a MORE ADVANCED version than Blueprint v1 assumed. Many items listed as "MISSING" in the blueprint have already been implemented. However, new bugs were introduced in the implementation.

Area	Blueprint v1 Status	Actual voice_mode Status
Interview hooks (useTTS, useSTT, etc.)	❌ MISSING — create all 4	✅ All 4 hooks EXIST (.js files)
Interview components (BrowserCheck, etc.)	❌ MISSING — create all 7	✅ All 7 components EXIST (.jsx files)
InterviewResults page	❌ MISSING	✅ EXISTS (InterviewResults.jsx)
submit-all endpoint	❌ MISSING — must create	✅ EXISTS in views.py
/services/openai_service.py	Not mentioned	✅ EXISTS — wraps Gemini, generates 8 questions + evaluate_interview()
Interview.jsx (state machine)	❌ MISSING	✅ EXISTS with 12-phase state machine
pdfGenerator.js	❌ MISSING	✅ EXISTS — html2canvas + jsPDF
browserSupport.js	❌ MISSING	⚠️ EXISTS but is EMPTY — 0 bytes
App.tsx routing	Partially defined	⚠️ EXISTS but has ROUTING CONFLICT (2 interview routes)
interviewAPI.ts (old)	⚠️ Hardcoded URL bug	⚠️ STILL has hardcoded localhost:8000

1.2 Critical Architecture Issue (NEW — Not in Blueprint v1)

🚨 DUAL INTERVIEW SYSTEM CONFLICT: There are TWO separate interview implementations in the codebase that are INCOMPATIBLE with each other. They share no code, use different APIs, and both are mounted in App.tsx. This will cause confusion and bugs in production.

Property	System A: InterviewSession.tsx	System B: Interview.jsx (Voice Mode)
File	pages/InterviewSession.tsx	pages/Interview.jsx
Route	/ai-interview-session	/interview
Architecture	Simple 5-phase state, inline fetch()	12-phase state machine with 4 custom hooks
TTS	Inline speakText() — NO Chrome fix	useTTS hook — Chrome fix INCLUDED
STT	Inline SpeechRecognition — no auto-restart	useSTT hook — auto-restart + en-IN locale
Silence detection	NONE — manual only	useSilenceDetector — 3s auto-advance

Property	System A: InterviewSession.tsx	System B: Interview.jsx (Voice Mode)
LocalStorage backup	NONE — refresh = lost session	useInterviewSession — full persistence
API URL	HARDCODED http://localhost:8000	Uses apiClient (env-var based)
Evaluation	POST /submit-all/ via fetch()	POST /submit-all/ via apiClient
Quality	Has 8 bugs — should be REPLACED	Industry-level — should be the PRIMARY system

⚡ DECISION: InterviewSession.tsx must be DELETED. Interview.jsx + its 4 hooks must become the ONLY interview system. All routing and navigation must point to /interview.

2. COMPLETE BUG INVENTORY — ALL ISSUES FOUND IN CODE

Every bug below was confirmed by reading the actual source files in the voice_mode ZIP. File paths and exact line references are provided for each issue.

2.1 CRITICAL Bugs (Will Break Core Functionality)

BUG-C1: Dual Interview Systems — Routing Conflict

Property	Detail
Severity	🔴 CRITICAL
Files	App.tsx (lines 11–14, 44–55), pages/InterviewSession.tsx, pages/Interview.jsx
Description	Two incompatible interview implementations are both mounted. /ai-interview-session uses old buggy system. /interview uses new voice-mode system. AIInterviewLanding.tsx navigates to /ai-interview-session (old broken system), NOT to /interview (new system). Users will never reach the good code.
Impact	All users directed to broken system. Voice hooks, silence detection, localStorage backup — all bypassed.
Fix	Step 1: Delete InterviewSession.tsx. Step 2: Update AIInterviewLanding.tsx navigate() call from /ai-interview-session to /interview. Step 3: Pass resumeId via state, not localStorage.

BUG-C2: pdfGenerator.js — Wrong Function Signature

Property	Detail
Severity	🔴 CRITICAL
File	frontend/src/utils/pdfGenerator.js (function signature)
Description	generateInterviewPDF() expects (candidateName, sessionDate) as parameters. But InterviewResults.jsx calls it as generateInterviewPDF(results) — passing the full results object as the first argument. The function then does results.replace() to sanitize the filename, which crashes because results is an object not a string.
Exact Bug	pdfGenerator.js: export async function generateInterviewPDF(candidateName = "Candidate", sessionDate = ...)
Called as	InterviewResults.jsx: const handlePDF = () => generateInterviewPDF(results);
Impact	PDF download crashes with TypeError every time. Cannot download any report.
Fix	Update pdfGenerator.js to accept (results) object. Extract name from results.candidate_name or use "Candidate". Or update InterviewResults.jsx to pass (results.candidate_name, results.date) explicitly.

BUG-C3: views.py — Wrong Import Path for openai_service

Property	Detail
Severity	🔴 CRITICAL

Property	Detail
File	backend/apps/interview/views.py (line 17)
Description	views.py imports: from services.openai_service import generate_questions, evaluate_interview. But the actual file is at backend/services/openai_service.py — which is outside the apps/ directory. Django's INSTALLED_APPS does not include a "services" app. This import will fail with ModuleNotFoundError on every server start.
Impact	Backend server fails to start. 0% of interview functionality works.
Fix	Option A: Move openai_service.py into apps/interview/services/ and fix import to: from .services.openai_service import generate_questions, evaluate_interview. Option B: Add backend/ to sys.path in settings.py (hacky — avoid). Option C: Create backend/services/__init__.py to make it a package and add to PYTHONPATH (cleanest for shared utils).

BUG-C4: google-generativeai Version Too Old

Property	Detail
Severity	🔴 CRITICAL
File	backend/requirements.txt (line: google-generativeai==0.3.0)
Description	requirements.txt pins google-generativeai==0.3.0. The openai_service.py uses genai.GenerativeModel() with system_instruction= and generation_config=genai.types.GenerationConfig() — these were added in v0.5+. With v0.3.0 installed, every Gemini call crashes with AttributeError.
Impact	Question generation and evaluation both fail. Interview cannot proceed.
Fix	Update requirements.txt: google-generativeai>=0.8.0. Run: pip install --upgrade google-generativeai. Verify: python -c "import google.generativeai as genai; print(genai.__version__)"

BUG-C5: browserSupport.js — Completely Empty File

Property	Detail
Severity	🔴 CRITICAL
File	frontend/src/utils/browserSupport.js
Description	The file exists but contains ZERO code (only a comment header). BrowserCheck.jsx imports and calls checkBrowserSupport() from this file. The function does not exist — this will throw a ReferenceError when BrowserCheck renders, crashing the entire Interview flow at phase 1.
Impact	Interview never starts. Crashes immediately at BrowserCheck phase.
Fix	Implement browserSupport.js with at minimum: checkBrowserSupport() function that returns { hasSpeechRecognition, hasSpeechSynthesis, hasMediaDevices, browser, isChrome, isEdge }. Full implementation provided in Section 4.2.

BUG-C6: InterviewSession.tsx — Hardcoded localhost URLs

Property	Detail
Severity	🔴 CRITICAL (in InterviewSession.tsx) / Already fixed in Interview.jsx
File	frontend/src/pages/InterviewSession.tsx (lines: apiPost calls)

Property	Detail
Description	InterviewSession.tsx uses apiPost('http://localhost:8000/api/interview/start/', ...) and apiPost('http://localhost:8000/api/interview/submit-all/', ...). Hardcoded localhost will break in ANY non-local environment. Note: Interview.jsx correctly uses apiClient from api.ts which reads VITE_API_BASE_URL.
Impact	Works only on developer machine. Breaks on deployment, staging, or any team member machine with different port.
Fix	This bug is moot if BUG-C1 fix (delete InterviewSession.tsx) is applied. If keeping InterviewSession.tsx for any reason, replace hardcoded URLs with import.meta.env.VITE_API_BASE_URL.

2.2 HIGH Severity Bugs (Degrade UX Significantly)

BUG-H1: STT Language Set to en-US, Should be en-IN

Property	Detail
Severity	🟠 HIGH
File	frontend/src/pages/InterviewSession.tsx (line: recognition.lang = 'en-US')
Description	InterviewSession.tsx sets recognition.lang = "en-US". The useSTT hook in hooks/useSTT.js correctly sets recognition.lang = "en-IN". Since InterviewSession.tsx will be deleted (BUG-C1 fix), this issue exists only in the old system. However, it is worth noting for any future reference. Indian-accented English is recognized significantly better with en-IN.
Impact	High transcription error rate for Indian-accented English users.
Fix	Ensure useSTT.js always uses en-IN. Add a comment explaining why en-IN is used.

BUG-H2: App.tsx — /interview Route Uses localStorage for resumeId

Property	Detail
Severity	🟠 HIGH
File	frontend/src/App.tsx (InterviewWrapper function)
Description	InterviewWrapper reads resumeId from localStorage: const resumeId = localStorage.getItem("resume_id") || "". This is fragile — localStorage key names are magic strings, prone to typos, and localStorage is not type-safe. If the user has multiple resumes or if the key is stale, wrong resume context is used for question generation.
Impact	Questions generated based on wrong or stale resume. Silent failure — no error shown.
Fix	Pass resumeId via React Router state: navigate('/interview', { state: { resumeId } }). In Interview.jsx: const { resumeId } = useLocation().state || {}. This is type-safe, scoped to navigation, and automatically cleared.

BUG-H3: useTTS.js — onvoiceschanged Does Not Actually Wait

Property	Detail
Severity	🟠 HIGH

Property	Detail
File	frontend/src/hooks/useTTS.js (useEffect, lines ~40-50)
Description	The hook registers onvoiceschanged but the speak() function does not await voice loading. If speak() is called within the first 100ms of component mount (before Chrome loads voices asynchronously), getVoice() returns null and the utterance plays in the browser's robotic default voice. The real fix requires speak() to wait for voices to be populated.
Impact	First question of every interview uses robotic default voice instead of natural Google voice.
Fix	In speak(): before creating the utterance, check if voices are loaded. If not, wrap in a Promise that resolves on onvoiceschanged. See Section 4.3 for complete fix.

BUG-H4: useSTT.js — Recognition Instance Created on Mount Only

Property	Detail
Severity	🟠 HIGH
File	frontend/src/hooks/useSTT.js (useEffect with empty deps)
Description	The SpeechRecognition instance is created once in useEffect([], []). If the user denies mic permission, then grants it (after going to browser settings), the hook's recognition instance is in an error state and cannot be reset without a full page reload. No retry mechanism exists.
Impact	Users who initially deny mic access, then grant it, cannot use voice input without refreshing the page.
Fix	Add a resetRecognition() function that tears down the existing instance and creates a new one. Call it from Interview.jsx when MicPermission is re-granted.

2.3 MEDIUM Severity Bugs

BUG-M1: evaluate_interview() — question_results Use question_index Not questionId

Property	Detail
Severity	🟡 MEDIUM
Files	backend/services/openai_service.py (prompt), backend/apps/interview/views.py (submit_all)
Description	The Gemini prompt for evaluate_interview() asks for question_results with question_index (1-8). views.py submit_all() then does: session.questions.get(question_number=q_index) — but the InterviewAnswer save also tries to set answer_text via question.answer_text which does NOT exist on InterviewQuestion model (answer_text is on InterviewAnswer, not InterviewQuestion). This causes an AttributeError when saving per-question results.
Impact	Per-question scores saved with AttributeError — answer text is not recorded correctly in DB.
Fix	In submit_all() view, change: answer_text: question.answer_text to answer_text: next(a["answerText"] for a in answers if ...). Match answer by question_number, not try to read it from question object.

BUG-M2: useSilenceDetector — handleSilence Stale Closure

Property	Detail
Severity	🟡 MEDIUM
File	frontend/src/pages/Interview.jsx (useSilenceDetector call and handleSilence function)
Description	useSilenceDetector(handleSilence, 3000) is called with handleSilence as the callback. handleSilence checks if (phase === PHASES.RECORDING). But useSilenceDetector creates a closure over handleSilence at render time. When the silence timer fires, it calls the STALE version of handleSilence that was captured when useSilenceDetector was first called — which has the initial phase value (BROWSER_CHECK), not RECORDING. The phase check always fails.
Impact	Silence detection never auto-advances. User must always manually click Skip. Core UX feature broken.
Fix	Use a ref to track phase inside handleSilence: const phaseRef = useRef(phase); useEffect(() => { phaseRef.current = phase; }, [phase]); Then in handleSilence: if (phaseRef.current === PHASES.RECORDING) finalizeAnswer();

BUG-M3: ReportTemplate.jsx Not Checked

Property	Detail
Severity	🟡 MEDIUM
File	frontend/src/components/results/ReportTemplate.jsx
Description	ReportTemplate is used by pdfGenerator.js to capture a hidden div for PDF generation. The element must have id="interview-report-pdf". If ReportTemplate does not render this id, the PDF generator logs an error and produces nothing. This was not verified in the code review as the file's internal structure was not examined in detail.
Impact	PDF generation silently fails if id is missing.
Fix	Verify ReportTemplate.jsx root element has id="interview-report-pdf". If not, add it.

2.4 LOW Severity Bugs

BUG-L1: authStore.ts Stores Only Access Token

Property	Detail
Severity	🟢 LOW (inherited from Blueprint v1 BUG-12)
File	frontend/src/store/authStore.ts
Description	authStore.ts stores only access_token. api.ts reads token directly from localStorage. These are inconsistent sources of truth. Not interview-specific but affects the entire app including interview API calls.
Impact	Token inconsistency can cause silent auth failures after token expiry.
Fix	Store both tokens in Zustand store OR both in localStorage — not split across both systems.

2.5 Bug Summary Table — Quick Reference

ID	File	Issue	Severity	Priority
BUG-C1	App.tsx + InterviewSession.tsx	Dual interview systems — routing conflict	🔴 CRITICAL	Fix first
BUG-C2	pdfGenerator.js	Wrong function signature — PDF always crashes	🔴 CRITICAL	Fix first
BUG-C3	views.py	Wrong import path for openai_service.py	🔴 CRITICAL	Fix first
BUG-C4	requirements.txt	google-generativeai v0.3.0 — too old	🔴 CRITICAL	Fix first
BUG-C5	browserSupport.js	Empty file — import crashes BrowserCheck	🔴 CRITICAL	Fix first
BUG-C6	InterviewSession.tsx	Hardcoded localhost URLs (moot if C1 fixed)	🔴 CRITICAL	Fix via C1
BUG-H1	InterviewSession.tsx	STT lang=en-US instead of en-IN	🟠 HIGH	Fix via C1
BUG-H2	App.tsx	resumeId from localStorage — fragile	🟠 HIGH	Week 1
BUG-H3	useTTS.js	speak() called before voices loaded	🟠 HIGH	Week 1
BUG-H4	useSTT.js	No retry after mic permission denied+granted	🟠 HIGH	Week 2
BUG-M1	views.py + openai_service.py	answer_text read from wrong object in submit_all	🟡 MEDIUM	Week 1
BUG-M2	Interview.jsx	handleSilence stale closure — silence never fires	🟡 MEDIUM	Week 1
BUG-M3	ReportTemplate.jsx	Verify id="interview-report-pdf" exists	🟡 MEDIUM	Week 2
BUG-L1	authStore.ts	Inconsistent token storage	🟢 LOW	Week 3

3. CORRECT ARCHITECTURE — TARGET STATE

After applying all fixes, the interview module should have exactly this architecture. No dual systems, no fragile data passing, no stale closures.

3.1 Correct File Map — Interview Module Only

```
frontend/src/
├── pages/
│   ├── Interview.jsx              ✅ KEEP — primary interview controller (state
machine)
│   ├── InterviewSession.tsx       ❌ DELETE — replaced entirely by Interview.jsx
│   ├── InterviewResults.jsx       ✅ KEEP — results dashboard
│   └── InterviewFeedback.tsx      ⚠️  KEEP but fix navigation (should show after
results)
│
├── components/interview/
│   ├── BrowserCheck.jsx           ✅ KEEP
│   ├── MicPermission.jsx          ✅ KEEP
│   ├── LoadingScreen.jsx          ✅ KEEP
│   ├── PreBrief.jsx               ✅ KEEP
│   ├── QuestionCard.jsx           ✅ KEEP
│   ├── AnswerReview.jsx           ✅ KEEP
│   └── StatusBadge.jsx            ✅ KEEP
│
├── components/results/
│   ├── ScoreGauge.jsx             ✅ KEEP
│   ├── QuestionFeedbackCard.jsx   ✅ KEEP
│   └── ReportTemplate.jsx         ⚠️  VERIFY id="interview-report-pdf" exists
│
├── hooks/
│   ├── useTTS.js                  ✅ KEEP — fix BUG-H3 (await voice load)
│   ├── useSTT.js                  ✅ KEEP — fix BUG-H4 (add resetRecognition)
│   ├── useSilenceDetector.js      ✅ KEEP — bug is in CALLER (Interview.jsx BUG-M2)
│   └── useInterviewSession.js     ✅ KEEP — no changes needed
│
├── services/
│   ├── api.ts                     ✅ KEEP — apiClient with VITE_API_BASE_URL
│   ├── api/interview.js           ✅ KEEP — startSession + submitAll
│   └── interviewAPI.ts            ⚠️  KEEP but do NOT use in Interview flow
│                                      (only used by old pages, will be removed)
│
└── utils/
    ├── pdfGenerator.js            ⚠️  FIX — update function signature (BUG-C2)
    └── browserSupport.js          ❌ IMPLEMENT — currently empty (BUG-C5)
```

3.2 Correct Data Flow — Interview Session

```
USER JOURNEY (correct after all fixes):

1. User goes to /ai-interview → AIInterviewLanding.tsx
2. User uploads/selects resume → navigate('/interview', { state: { resumeId } })
3. App.tsx: <Route path="/interview" element={<ProtectedRoute><InterviewWrapper
/></ProtectedRoute>} />
4. InterviewWrapper reads resumeId from location.state (NOT localStorage)
5. Interview.jsx mounts → BROWSER_CHECK phase
6. BrowserCheck.jsx calls checkBrowserSupport() from browserSupport.js
7. MicPermission.jsx requests mic → startInterview() called
8. startSession(resumeId) → POST /api/interview/start/ → 8 questions returned
9. PRE_BRIEF → user clicks Begin → runQuestion(0)
```

```
10. SPEAKING → useTTS.speak() → awaits voice load → reads question
11. COUNTDOWN → 2 seconds
12. RECORDING → useSTT.startRecording() + useSilenceDetector.start()
13. phaseRef (BUG-M2 fix) detects phase = RECORDING → silence fires → finalizeAnswer()
14. useInterviewSession.saveAnswer() → localStorage backup
15. Repeat for questions 1-7
16. REVIEW → AnswerReview.jsx → user reviews/re-records
17. SUBMITTING → submitAll(sessionId, answers) → POST /api/interview/submit-all/
18. Backend: evaluate_interview(answers) → Gemini single call → evaluation JSON
19. RESULTS → InterviewResults.jsx → scores, feedback, PDF download
20. handlePDF → generateInterviewPDF(results) — correct signature after BUG-C2 fix
```

4. STEP-BY-STEP FIX IMPLEMENTATION

Each fix is self-contained. Apply them in the order shown — earlier fixes are prerequisites for later ones. Each step includes the exact code change needed.

4.1 FIX: BUG-C3 — Move openai_service.py (Backend — Fix First)

🔴 This must be done before any other backend fix. Nothing works until the import error is resolved.

Step 1: Move the file

```
# From terminal in the backend directory:
mv backend/services/openai_service.py backend/apps/interview/services/openai_service.py

# OR on Windows:
move backend\services\openai_service.py backend\apps\interview\services\
openai_service.py
```

Step 2: Fix the import in views.py

```
# In backend/apps/interview/views.py — FIND this line (around line 17):
from services.openai_service import generate_questions, evaluate_interview

# REPLACE with:
from .services.openai_service import generate_questions, evaluate_interview
```

Step 3: Create __init__.py if missing

```
# Make sure this file exists (it probably does, but verify):
backend/apps/interview/services/__init__.py

# If missing, create it (can be empty):
touch backend/apps/interview/services/__init__.py
```

Step 4: Verify the fix

```
cd backend
python manage.py check --deploy
# Should show 0 errors related to interview module
```

4.2 FIX: BUG-C4 — Update google-generativeai Version

```
# In backend/requirements.txt — FIND:
google-generativeai==0.3.0

# REPLACE WITH:
google-generativeai>=0.8.0

# Then run:
pip install --upgrade google-generativeai

# Verify:
python -c "import google.generativeai as genai; print(genai.__version__)"
# Should print 0.8.x or higher
```

4.3 FIX: BUG-C5 — Implement browserSupport.js

The file exists but is empty. Replace its contents entirely with this implementation:

```
// frontend/src/utils/browserSupport.js
// Browser compatibility detection for the AI Interview module

/**
 * checkBrowserSupport()
 * Returns an object describing what voice/media features this browser supports.
 * Called by BrowserCheck.jsx on phase entry.
 *
 * @returns {Object} {
 *   hasSpeechRecognition  {boolean},  // Web Speech API available
 *   hasSpeechSynthesis    {boolean},  // TTS available
 *   hasMediaDevices       {boolean},  // navigator.mediaDevices exists
 *   browser               {string},   // 'chrome' | 'edge' | 'safari' | 'firefox' |
'other'
 *   isChrome              {boolean},
 *   isEdge                {boolean},
 *   isSafari              {boolean},
 *   isFirefox             {boolean},
 *   recommendation        {string},   // human-readable suggestion
 * }
 */
export function checkBrowserSupport() {
  const ua = navigator.userAgent.toLowerCase();

  const isChrome  = /chrome/.test(ua)  && !/edg/.test(ua) && !/opr/.test(ua);
  const isEdge    = /edg//.test(ua);
  const isSafari  = /safari/.test(ua)  && !/chrome/.test(ua);
  const isFirefox = /firefox/.test(ua);

  const browser = isChrome  ? 'chrome'
                : isEdge    ? 'edge'
                : isSafari  ? 'safari'
                : isFirefox ? 'firefox'
                : 'other';

  const hasSpeechRecognition = (
    'SpeechRecognition' in window || 'webkitSpeechRecognition' in window
  );
  const hasSpeechSynthesis = 'speechSynthesis' in window;
  const hasMediaDevices    = !!(navigator.mediaDevices &&
navigator.mediaDevices.getUserMedia);

  let recommendation = '';
  if (!hasSpeechRecognition) {
    recommendation = 'Voice input is not supported. Please use Google Chrome or
Microsoft Edge for the best experience.';
  } else if (!hasSpeechSynthesis) {
    recommendation = 'Text-to-speech is not available. Questions will be displayed as
text only.';
  } else if (isFirefox) {
    recommendation = 'Firefox has limited Web Speech API support. Chrome or Edge
recommended for voice interviews.';
  } else if (isChrome || isEdge) {
    recommendation = 'Your browser fully supports voice interviews.';
  }

  return {
    hasSpeechRecognition,
    hasSpeechSynthesis,
    hasMediaDevices,
    browser,
    isChrome,
    isEdge,
```

```
    isSafari,
    isFirefox,
    recommendation,
  };
}

/**
 * isBrowserSupported()
 * Quick boolean check — returns true if the browser can run the voice interview.
 * Used by BrowserCheck.jsx to decide whether to show a warning.
 */
export function isBrowserSupported() {
  const support = checkBrowserSupport();
  return support.hasSpeechRecognition && support.hasMediaDevices;
}
```

4.4 FIX: BUG-C1 — Remove Dual Interview System

Step 1: Delete the old interview session page

```
# Delete the old system file:
rm frontend/src/pages/InterviewSession.tsx

# This file should NOT exist after this step.
```

Step 2: Update AIInterviewLanding.tsx navigation

```
// In frontend/src/pages/AIInterviewLanding.tsx
// FIND the navigate call (somewhere in the "Start Interview" button handler):
navigate('/ai-interview-session', { state: { config, resume } });
// OR it might be:
navigate('/ai-interview-session');

// REPLACE WITH:
navigate('/interview', { state: { resumeId: resume?.id || selectedResumeId } });
// Note: only pass resumeId, not the full resume object
```

Step 3: Update InterviewWrapper in App.tsx

```
// In frontend/src/App.tsx — FIND the InterviewWrapper function:
function InterviewWrapper() {
  const resumeId = localStorage.getItem('resume_id') || '';  // ← FRAGILE (BUG-H2)
  return <Interview resumeId={resumeId} />;
}

// REPLACE WITH:
import { useLocation } from 'react-router-dom';

function InterviewWrapper() {
  const location = useLocation();
  const resumeId = location.state?.resumeId || '';  // ← SAFE: from navigation state
  return <Interview resumeId={resumeId} />;
}

// Also REMOVE the /ai-interview-session route entirely:
// DELETE: <Route path="/ai-interview-session" element={...InterviewSessionPage...} />
// KEEP:   <Route path="/interview" element={<ProtectedRoute><InterviewWrapper
/></ProtectedRoute>} />
```

4.5 FIX: BUG-C2 — Fix pdfGenerator.js Function Signature

```
// In frontend/src/utils/pdfGenerator.js
// FIND the function signature:
export async function generateInterviewPDF(
  candidateName = 'Candidate',
  sessionDate   = new Date().toISOString().split('T')[0]
) {

// REPLACE WITH (accept results object, extract name internally):
export async function generateInterviewPDF(results = {}) {
  // Extract a safe candidate name from results (or use default)
  const candidateName = results.candidate_name
    || results.user_name
    || 'Candidate';
  const sessionDate = results.session_date
    || new Date().toISOString().split('T')[0];

// The rest of the function stays the same — but now it uses the
// locally extracted candidateName and sessionDate variables.
// The line: const safeName = candidateName.replace(...) will now work correctly.
```

4.6 FIX: BUG-M2 — Fix Stale Closure in Interview.jsx (Silence Detection)

> ⚡ This is the most subtle bug. Without this fix, silence detection never fires. The voice interview cannot auto-advance.

```
// In frontend/src/pages/Interview.jsx

// STEP 1: Add a phaseRef that always holds the current phase value.
// Add these lines right after the phase state declaration:

const [phase, setPhase] = useState(PHASES.BROWSER_CHECK);
// ADD THESE TWO LINES:
const phaseRef = useRef(PHASES.BROWSER_CHECK);
useEffect(() => { phaseRef.current = phase; }, [phase]);

// STEP 2: Update handleSilence to use phaseRef instead of the stale phase state.
// FIND the handleSilence function:
function handleSilence() {
  if (phase === PHASES.RECORDING) {   // ← STALE — always sees initial phase
    finalizeAnswer();
  }
}

// REPLACE WITH:
function handleSilence() {
  if (phaseRef.current === PHASES.RECORDING) {  // ← FRESH — always current
    finalizeAnswer();
  }
}

// No changes needed to useSilenceDetector.js — the bug was in the caller.
```

4.7 FIX: BUG-H3 — Fix useTTS.js Voice Loading Race Condition

```
// In frontend/src/hooks/useTTS.js
// REPLACE the speak() function with a version that waits for voices to load:

const speak = useCallback((text) => {
```

```
  return new Promise((resolve, reject) => {
    synth.current.cancel();

    // ── WAIT FOR VOICES TO BE LOADED (fixes BUG-H3) ──────────────────
    const doSpeak = () => {
      const utter = new SpeechSynthesisUtterance(text);
      utter.rate   = 0.88;
      utter.pitch  = 1.0;
      utter.volume = 1.0;

      const voice = getVoice();
      if (voice) utter.voice = voice;

      // Chrome keep-alive ping (BUG-06 fix from Blueprint — keep this)
      const ping = setInterval(() => {
        if (synth.current.paused) synth.current.resume();
      }, 5000);

      utter.onend  = () => { clearInterval(ping); resolve(); };
      utter.onerror = (err) => { clearInterval(ping); reject(err); };

      synth.current.speak(utter);
    };

    // If voices already loaded — speak immediately
    if (synth.current.getVoices().length > 0) {
      doSpeak();
    } else {
      // Wait for onvoiceschanged to fire, then speak
      synth.current.onvoiceschanged = () => {
        synth.current.onvoiceschanged = null; // clear handler
        doSpeak();
      };
      // Safety timeout: if voices never load in 2s, speak with default voice
      setTimeout(() => {
        if (synth.current.getVoices().length === 0) {
          synth.current.onvoiceschanged = null;
          doSpeak(); // speak anyway with default voice
        }
      }, 2000);
    }
  });
}, [getVoice]);
```

4.8 FIX: BUG-M1 — Fix answer_text in submit_all view

```
# In backend/apps/interview/views.py — FIND the submit_all() function.
# FIND the per-question results loop (around line: for qr in evaluation...):

for qr in evaluation.get('question_results', []):
    q_index = qr.get('question_index')
    if q_index is None:
        continue
    try:
        question = session.questions.get(question_number=q_index)
        InterviewAnswer.objects.update_or_create(
            question=question,
            defaults={
                'answer_text': question.answer_text or '[No answer provided]',  # ← BUG:
wrong
                ...
            }
        )
```

```
# REPLACE the answer_text line WITH:
# Build an index of answers from the request so we can look up the text
answers_by_index = {}
for i, ans in enumerate(answers, start=1):
    answers_by_index[i] = ans.get('answerText', ans.get('answer_text', '[No answer
provided]'))

# Then in the loop:
for qr in evaluation.get('question_results', []):
    q_index = qr.get('question_index')
    if q_index is None:
        continue
    try:
        question = session.questions.get(question_number=q_index)
        InterviewAnswer.objects.update_or_create(
            question=question,
            defaults={
                'answer_text': answers_by_index.get(q_index, '[No answer provided]'),  #
← FIXED
                'score':        to_100(qr.get('score')),
                'ai_feedback':  qr.get('feedback', ''),
                'strengths':    [qr.get('strength', '')] if qr.get('strength') else [],
                'improvements': [qr.get('improvement', '')] if qr.get('improvement')
else [],
            }
        )
    except InterviewQuestion.DoesNotExist:
        continue
```

5. TESTING PROTOCOL — VERIFY EACH FIX

After applying each fix, run the corresponding test. Do NOT move to the next fix until the current test passes. These are the minimum tests for each bug.

Bug ID	Test Steps	Expected Result	Status
BUG-C3	cd backend && python manage.py check	No errors printed. Server starts.	⬜ Todo
BUG-C4	python -c "import google.generativeai; print(google.generativeai.__version__)"	Prints 0.8.x or higher	⬜ Todo
BUG-C5	Open browser, go to /interview. BrowserCheck phase renders.	No "ReferenceError: checkBrowserSupport is not defined" in console	⬜ Todo
BUG-C1	Click Start Interview on /ai-interview landing page	Navigates to /interview (NOT /ai-interview-session). URL shows /interview.	⬜ Todo
BUG-C2	Complete an interview, click Download PDF	PDF file downloads. No TypeError in console.	⬜ Todo
BUG-M2	Answer a question, then stop speaking for 4 seconds	After ~3s of silence, question auto-advances without clicking Skip.	⬜ Todo
BUG-H3	Open a fresh incognito tab, go to /interview, let first question play	Question is read in a natural Google voice (not robotic default)	⬜ Todo
BUG-M1	Complete interview, check Django admin → InterviewAnswer records	Each answer has the actual answer text, not "[No answer provided]"	⬜ Todo
BUG-H2	Open two tabs. Select Resume A in tab 1, start interview. In tab 2 select Resume B. Back in tab 1, start interview.	Tab 1 interview uses Resume A. Not overwritten by tab 2.	⬜ Todo
ALL	Full end-to-end: register → upload resume → start interview → answer 8 questions → review → submit → see results → download PDF	All phases complete without error. PDF downloaded.	⬜ Todo

5.1 Postman / API Tests for Backend Fixes

```
# Test 1: Verify /api/interview/start/ works after BUG-C3 fix
POST http://localhost:8000/api/interview/start/
Headers: Authorization: Bearer <token>
Body: { "resume_id": "<valid_uuid>" }
Expected: 201 { "session_id": "...", "questions": [ {8 items} ] }

# Test 2: Verify /api/interview/submit-all/ saves answers correctly (BUG-M1 fix)
POST http://localhost:8000/api/interview/submit-all/
Headers: Authorization: Bearer <token>
Body: {
  "session_id": "<session_uuid>",
  "answers": [
    {"questionId": "...", "questionText": "Tell me about yourself",
```

```
     "questionType": "HR", "answerText": "I am a final year student..."},
    ... (8 total)
  ]
}
Expected: 200 { "overall_score": 7.2, "placement_readiness": "almost_ready", ... }

# After Test 2: Check Django Admin
# Go to: http://localhost:8000/admin/interview/interviewanswer/
# Each answer should show the actual text, NOT "[No answer provided]"
```

6. IMPLEMENTATION ROADMAP — SPRINT-BY-SPRINT

The roadmap is structured so that each sprint only builds on verified, working code from the previous sprint. Sprint 0 is mandatory before anything else.

Sprint	Duration	Focus	Key Deliverables	Success Criteria
Sprint 0	1–2 days	Critical backend unblock	Fix BUG-C3 (import path) Fix BUG-C4 (Gemini version) Fix BUG-M1 (answer_text) Verify /api/interview/start/ works in Postman	POST /start/ returns 8 questions. POST /submit-all/ saves answers correctly.
Sprint 1	2–3 days	Frontend critical fixes	Fix BUG-C5 (browserSupport.js) Fix BUG-C1 (delete InterviewSession.tsx) Update AIInterviewLanding.tsx routing Fix BUG-H2 (InterviewWrapper — use location.state) Fix BUG-M2 (phaseRef stale closure)	Full interview flow reachable via /interview. Silence detection auto-advances after 3s. No crashes at BrowserCheck phase.
Sprint 2	1–2 days	Voice quality fixes	Fix BUG-H3 (useTTS voice load wait) Fix BUG-H4 (useSTT reset after mic retry) Fix BUG-C2 (pdfGenerator signature) Verify PDF download works end-to-end	First question reads in Google voice. PDF downloads without TypeError. Mic retry after permission re-grant works.
Sprint 3	1 day	Polish and validation	Fix BUG-L1 (authStore token consistency) Verify ReportTemplate has correct PDF id Full end-to-end test with real resume Check all 14 bugs marked resolved	Full flow: register → resume → interview → PDF All 14 bugs show ✅ resolved in checklist.

6.1 Day-by-Day Sprint 0 Tasks (Most Critical)

These are the exact commands and code changes for Sprint 0 in order:

1. Open requirements.txt. Change google-generativeai==0.3.0 to google-generativeai>=0.8.0. Save.
2. Run: pip install -r requirements.txt
3. Move openai_service.py: mv backend/services/openai_service.py backend/apps/interview/services/openai_service.py
4. Edit views.py line 17: change from services.openai_service to from .services.openai_service
5. Run: python manage.py check — verify 0 errors
6. Test in Postman: POST /api/interview/start/ with valid JWT and resume_id
7. Confirm 8 questions returned in response
8. Fix BUG-M1 in views.py submit_all() (see Section 4.8)
9. Test: POST /api/interview/submit-all/ with mock 8 answers
10. Verify Django Admin shows correct answer text for each InterviewAnswer record

7. ENVIRONMENT SETUP & CONFIGURATION

7.1 Backend .env File

⚠️ Create this file at: backend/.env — Do NOT commit to Git. Add to .gitignore.

```
# backend/.env

# Django
SECRET_KEY=your-long-random-secret-key-here-at-least-50-chars
DEBUG=True
ALLOWED_HOSTS=localhost,127.0.0.1

# Database
DATABASE_URL=sqlite:///db.sqlite3

# REQUIRED: Get a free key from https://aistudio.google.com/
GEMINI_API_KEY=AIzaSy...your-actual-key-here...

# JWT tokens
JWT_ACCESS_TOKEN_LIFETIME=60     # minutes
JWT_REFRESH_TOKEN_LIFETIME=7     # days

# CORS — allow React dev server
CORS_ORIGINS=http://localhost:5173,http://localhost:3000
```

7.2 Frontend .env File

ℹ️ Create this file at: frontend/.env — This is the BUG-11 fix from Blueprint v1.

```
# frontend/.env

# Backend API base URL — change for production deployment
VITE_API_BASE_URL=http://localhost:8000/api
```

7.3 Quick Start Commands

```
# ── BACKEND ──────────────────────────────────────────────────────────────
cd ai-trainer/backend

python -m venv venv
source venv/bin/activate       # Mac/Linux
# OR: venv\Scripts\activate   # Windows

pip install -r requirements.txt
pip install --upgrade google-generativeai   # Ensures >=0.8.0

# Create .env file with GEMINI_API_KEY (see Section 7.1)

python manage.py makemigrations
python manage.py migrate
python manage.py createsuperuser  # optional but useful for admin panel

python manage.py runserver 8000

# ── FRONTEND ─────────────────────────────────────────────────────────────
```

```
cd ai-trainer/frontend

npm install
# Create .env file (see Section 7.2)

npm run dev
# App available at: http://localhost:5173
```

8. FINAL BUG FIX CHECKLIST

Use this as your daily progress tracker. Mark each item ONLY after implementing AND testing the fix — not just when the code looks right.

Bug ID	File(s)	Fix Description	Test	Done?
BUG-C3	views.py	Fix import path for openai_service.py	python manage.py check	⬜
BUG-C4	requirements.txt	Update google-generativeai to >=0.8.0	Verify version number	⬜
BUG-M1	views.py submit_all()	Fix answer_text read from wrong object	Check DB in admin panel	⬜
BUG-C5	browserSupport.js	Implement checkBrowserSupport() function	/interview loads BrowserCheck	⬜
BUG-C1	App.tsx + AIInterviewLanding.tsx	Delete InterviewSession.tsx, fix routing to /interview	Start Interview navigates to /interview	⬜
BUG-H2	App.tsx InterviewWrapper	Read resumeId from location.state not localStorage	Check Network tab: start/ called with correct resume_id	⬜
BUG-M2	Interview.jsx	Add phaseRef, update handleSilence to use phaseRef	Silence for 4s auto-advances question	⬜
BUG-H3	useTTS.js speak()	Wait for voices loaded before speaking	First question reads in Google voice on fresh page load	⬜
BUG-H4	useSTT.js	Add resetRecognition() for retry after mic denied+granted	Deny mic, grant mic in settings, retry — voice works	⬜
BUG-C2	pdfGenerator.js	Fix function signature to accept (results) object	PDF downloads without TypeError	⬜
BUG-M3	ReportTemplate.jsx	Verify root element has id="interview-report-pdf"	PDF contains full report content	⬜
BUG-C6	InterviewSession.tsx	Resolved by BUG-C1 fix (file deleted)	N/A — resolved via C1	✅
BUG-H1	InterviewSession.tsx	Resolved by BUG-C1 fix (file deleted)	N/A — resolved via C1	✅
BUG-L1	authStore.ts	Unify token storage (both in Zustand or both in localStorage)	Refresh page after login — still logged in	⬜

🎯 When ALL checkboxes above are ticked, the AI Interview Module is production-grade and industry-ready.

— End of Implementation Guide —

AI Pre-Placement Trainer | AI Interview Module | Code Review March 2026
YSPM's Yashoda Technical Campus, Satara | AI & Data Science Department